

Statistics Basics

The Math Behind Every Data Insight

■ STATISTICAL THINKING

■ LEARNING OUTCOME

By the end of this PDF, you will understand the core statistical concepts used in data analytics: central tendency, spread, probability, distributions, and hypothesis testing.

■ Skills You Will Gain

- Calculate and interpret mean, median, mode
- Understand variance, standard deviation, IQR
- Read and interpret data distributions
- Understand correlation vs causation
- Apply basic hypothesis testing (t-test, p-value)
- Use probability for data-driven decisions

■ Statistics separates analysts from data entry clerks. With stats, you find insights others miss.

SECTION 1

Measures of Central Tendency

	Formula	When to Use
	Sum / Count → sensitive to outliers	Normal distribution, no extreme values
	Middle value when sorted → resistant to outliers	Skewed data, income, house prices
	Most frequent value	Categorical data, most popular item

```
import numpy as np
data = [10, 20, 20, 30, 40, 1000] # outlier = 1000
print('Mean:', np.mean(data)) # 186.7 – pulled up by outlier
print('Median:', np.median(data)) # 25.0 – not affected
from scipy import stats
print('Mode:', stats.mode(data).mode[0]) # 20
```

SECTION 2

Measures of Spread

Measure	What It Tells You	Formula/Code
Range	Max - Min value span	<code>np.max(d) - np.min(d)</code>
Variance	Average squared deviation from mean	<code>np.var(data)</code>
Std Deviation	Square root of variance — same unit as data	<code>np.std(data)</code>
IQR	Q3 - Q1 — spread of middle 50%	<code>from scipy.stats import iqr; iqr(data)</code>
CV (Coeff. of Variation)	Std/Mean × 100 — relative spread %	<code>np.std(d)/np.mean(d)*100</code>

SECTION 3

Data Distributions

	Shape	Real-World Example
Normal (Symmetric)	Symmetric, mean=median=mode	Height, test scores, measurement errors
Right-Skewed (Long tail to the right)	Long tail to the right	Income, house prices (few very rich)
Left-Skewed (Long tail to the left)	Long tail to the left	Age at retirement, final exam scores
Uniform	All values equally likely	Rolling a dice, random number generator
Bimodal	Two peaks	Customer visit time (lunch + dinner rush)

68% of data falls within 1 std dev of mean
95% falls within 2 std devs
99.7% falls within 3 std devs
Use this to identify outliers: values beyond 3 std devs are unusual.

Correlation

```
# Pearson Correlation Coefficient (-1 to +1) import pandas as pd
corr = df[['StudyHours', 'ExamScore']].corr()
print(corr) # +1 = perfect positive, -1 = perfect negative, 0 = no correlation
# Correlation matrix for all numeric columns df.corr()
```

Ice cream sales and drowning rates are correlated — both increase in summer. But eating ice cream does NOT cause drowning. The hidden variable is HOT WEATHER. Always think about confounding variables before claiming causation.

Hypothesis Testing

1	State H0 & H1 H0 (null): 'New drug has no effect' H1 (alternative): 'New drug reduces fever'
2	Choose significance $\alpha = 0.05$ (5% significance level — industry standard)
3	Run test t-test for means, chi-square for categories, ANOVA for 3+ groups
4	Get p-value p-value = probability of seeing this result if H0 is true
5	Decide $p < 0.05 \rightarrow$ Reject H0 (result is statistically significant) $p > 0.05 \rightarrow$ Fail to reject H0

```
from scipy import stats # One-sample t-test
t_stat, p_value = stats.ttest_1samp(data, popmean=50)
print(f'p-value: {p_value:.4f}') # <0.05 means significant
```

■ PRACTICE QUESTIONS

Q1. Why is median preferred over mean for salary data?

Answer: Salary is right-skewed (a few very high earners). The median is not affected by extreme values, so it better represents the 'typical' salary.

Q2. What does a correlation of -0.85 between rain and sales mean?

Answer: A strong negative correlation — as rainfall increases, sales decrease significantly.

Q3. If p-value = 0.03 and $\alpha = 0.05$, what do you conclude?

Answer: $p < \alpha$, so we reject the null hypothesis. The result is statistically significant.

Q4. What does standard deviation measure?

Answer: How spread out the values are around the mean. High std dev = widely spread; low = tightly clustered.

Q5. A dataset has mean=50 and std=10. Is a value of 85 an outlier?

Answer: Yes — 85 is 3.5 std devs from the mean ($50 + 3 \times 10 = 80$). Values beyond 3 std devs are considered outliers.

■ Statistics Cheat Sheet	
<code>np.mean(data)</code>	Calculate mean
<code>np.median(data)</code>	Calculate median
<code>np.std(data)</code>	Standard deviation
<code>np.var(data)</code>	Variance
<code>df.corr()</code>	Correlation matrix
<code>stats.ttest_1samp(d, mu)</code>	One-sample t-test
<code>stats.ttest_ind(a, b)</code>	Two-sample t-test
<code>stats.chi2_contingency(t)</code>	Chi-square test
<code>p < 0.05</code>	Result is significant
<code>Z = (x-mean)/std</code>	Z-score formula
<code>IQR = Q3 - Q1</code>	Interquartile range
<code>Outlier if Z > 3</code>	Outlier detection rule